

DataSage AI Report

Dataset Overview

DATASET OVERVIEW

Rows: 100
Columns: 10
Missing Values: 0
Duplicate Rows: 0

DESCRIPTIVE STATISTICS

Units_Sold
Average: 5128.71
Minimum: 124.00
Maximum: 9925.00

Unit_Price
Average: 276.76
Minimum: 9.33
Maximum: 668.27

KPI Columns:
- Units_Sold
- Unit_Price

Category Columns:
- Region
- Country
- Item_Type
- Sales_Channel
- Order_Priority

Date Column: Ship_Date

Most Common Region: Sub-Saharan Africa
Most Common Country: The Gambia

CORRELATION ANALYSIS

Strongest Negative: Order_ID ↔ Units_Sold (-0.22)

AI Executive Summary

Executive Summary

- The dataset contains 100 rows and 10 columns, with no missing values and no duplicate rows.
- Key numeric KPIs provided are Units_Sold (average 5,128.71; min 124; max 9,925) and Unit_Price (average 276.76; min 9.33; max 668.27).
- Categorical fields include Region, Country, Item_Type, Sales_Channel, and Order_Priority; date is captured in Ship_Date.
- The most common Region in the sample is Sub-Saharan Africa and the most common Country is

The Gambia.

- Correlation analysis shows the strongest negative relationship is between Order_ID and Units_Sold (correlation = -0.22). This is the strongest negative correlation reported.

Key Findings

- Data quality is high:
- 0 missing values and 0 duplicate rows across 100 records.
- KPI summaries:
- Units_Sold: mean = 5,128.71; observed range [124, 9,925].
- Unit_Price: mean = 276.76; observed range [9.33, 668.27].
- Category coverage:
- Categorical fields available for segmentation: Region, Country, Item_Type, Sales_Channel, Order_Priority.
- Temporal field available for trend analysis: Ship_Date.
- Modal values:
- Most common Region: Sub-Saharan Africa.
- Most common Country: The Gambia.
- Correlation:
- The strongest reported negative correlation is Order_ID ↔ Units_Sold at -0.22.
- A correlation of -0.22 indicates a weak negative relationship; it is the strongest negative association reported but is not large in magnitude.

Business Recommendations

- Do not use Order_ID as a predictor in models:
- The negative correlation (-0.22) is weak and Order_ID is likely an identifier rather than a causal feature.
- Explore pricing and volume dynamics:
- Given the wide Unit_Price range (9.33 to 668.27) and wide Units_Sold range, run analyses by Item_Type and Sales_Channel to understand price elasticity and volume drivers.
- Leverage categorical and date fields for segmentation and trend detection:
- Segment KPIs by Region, Country, Item_Type, Sales_Channel, and Order_Priority.
- Use Ship_Date to analyze seasonality or shipment trends over time.
- Validate high-frequency geography:
- Investigate why Sub-Saharan Africa and The Gambia are the most common region/country in this sample to confirm whether this reflects business focus or sampling bias.

Risks

- Representativeness:
- The dataset is limited to 100 rows; findings may not generalize beyond this sample.
- Misuse of identifiers:
- Treating Order_ID as a numeric predictor could produce misleading model behavior; the observed correlation does not imply causation.
- Heterogeneity of items and prices:
- Wide Unit_Price and Units_Sold ranges suggest heterogeneous item types or markets; aggregating without segmentation could obscure important patterns.
- Overinterpretation of modal values:
- The most common Region/Country in the sample does not necessarily indicate highest sales or profitability without further aggregation and context.

Next Steps

- Compute derived metrics:
- Calculate revenue per order (Units_Sold × Unit_Price) and analyze total and average revenue by category.
- Segment and drill down:
- Analyze KPIs by Item_Type, Sales_Channel, Region, Country, and Order_Priority to identify high- and low-performing segments.
- Time-series analysis:

- Use Ship_Date to examine trends, seasonality, and changes in Units_Sold and Unit_Price over time.
- Correlation and feature assessment:
 - Build a full correlation matrix across numeric variables and assess which features have meaningful relationships with revenue and Units_Sold.
- Outlier investigation:
 - Inspect orders at the extremes of Units_Sold and Unit_Price for data entry issues or special cases that warrant separate handling.
- Data augmentation (if needed):
 - If broader inference is required, collect additional records or external context (market or demographic data) to improve representativeness.

Forecast Summary

Executive Summary

- Forecast for Units_Sold over the next 30 days predicts a growth of 0.02%.
- This magnitude indicates effectively flat demand for the forecast period.

Key Findings

- Target KPI: Units_Sold.
- Forecast period: 30 days.
- Predicted growth: 0.02% (near-zero change).
- Implication: expected unit demand is stable with minimal change versus the baseline.

Business Recommendations

- Maintain current inventory and supply commitments; avoid large adjustments based solely on this forecast.
- Monitor daily sales and report any deviation promptly to detect early signals.
- Run small, targeted marketing or pricing tests if the objective is to lift demand slightly; evaluate impact before scaling.
- Prioritize cost control and operational efficiency given limited near-term volume upside.

Risks

- A 0.02% growth forecast is close to noise—overreacting (e.g., increasing orders or cutting investment) could create inventory or service issues.
- External shocks or short-term promotions could move actual sales away from this flat projection.
- If baseline volume is very low, percent changes may be misleading; interpret in context of absolute units.

Next Steps

- Establish daily monitoring and a 7-day checkpoint to validate the forecast against actual Units_Sold.
- If improvement is needed, run controlled marketing tests and reassess after one full reporting cycle.
- Update the forecast promptly as new sales data or market signals arrive.

Generated by DataSage AI